MILITARY SPECIFICATION
IGNITER, AUXILIARY

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an auxiliary igniter for use in aircraft ejection seat catapults.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-1876	Aluminum Foil
TT-I-1795	Ink, Marking Stencil, Opaque (Porous and Non-Porous Surfaces)
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation-Packaging, Methods of

Naval Air Systems Command (Code Ident 30003)

AS 2683	Ignition Material, AIVC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commanding Officer, Naval Ordnance Station (611), Indian Head, Md.) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods
MIL-STD-1166	Radiographic Testing Requirements for Solid Propellants
MIL-STD-1167	Ammunition Data Card

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

LD 255173	Signal Container Mk 3 Mod 0 (for 30 Igniters)

Naval Ordnance Station, Indian Head, Md (Code Ident 14083)

8596874	Igniter Auxiliary Assembly
8596875	Case, Igniter Auxiliary

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations

49 CFR 171-190	Transportation

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1975 Edition and Revisions) available from the Superintendent of Documents, Government Printing Office, Washington, D.C. Order for the above publication should cite "49 CPR 100-199 (lastest revision)").

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified, a preproduction sample consisting of 40 igniters shall be delivered to the Government activity specified in the contract or purchase order (see 6.2) for testing. The sample shall be produced using methods and equipment proposed for the production lot. It shall be the responsibility of the contractor to inform the Government procuring activity of proposed changes after approval of the preproduction sample. In the event of significant changes in the process methods and equipment which in the opinion of the Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

procuring activity may adversely affect the characteristics of the material, additional preproduction samples may be required. Any production of the igniter prior to approval by the Government procuring activity of the preproduction lot shall be at the supplier's risk.

3.2 Conformance to documents. Unless otherwise specified, the igniter shall conform to the requirements specified herein and on Drawings 8596874 and 8596875.

3.3 Data. A certified analysis for each lot of ignition material shall be submitted to the procuring activity in accordance with 4.6. Data cards shall also be provided in accordance with 5.4.

3.4 Materials. All materials and parts used in the igniters shall be of the highest quality and functionally suitable for the intended application.

3.5 Weight. The contractor shall provide the igniter case weight, to within ±0.001 gram, with each loaded igniter. The weight of the loaded igniter shall conform to Drawing 8596874. The change in weight of each loaded igniter shall not be greater than 0.010 gram when subjected to environmental testing in accordance with 4.5.1.

3.6 Discontinuities and voids. There shall be no crack, void, or irregularity in the igniter longer than 1/8 inch or wider than 1/64 inch. There shall be no charge material beyond the outer edge of the igniter case (Figure 1). The supplier shall submit one set of radiographs to the Government procuring activity for examination and approval.

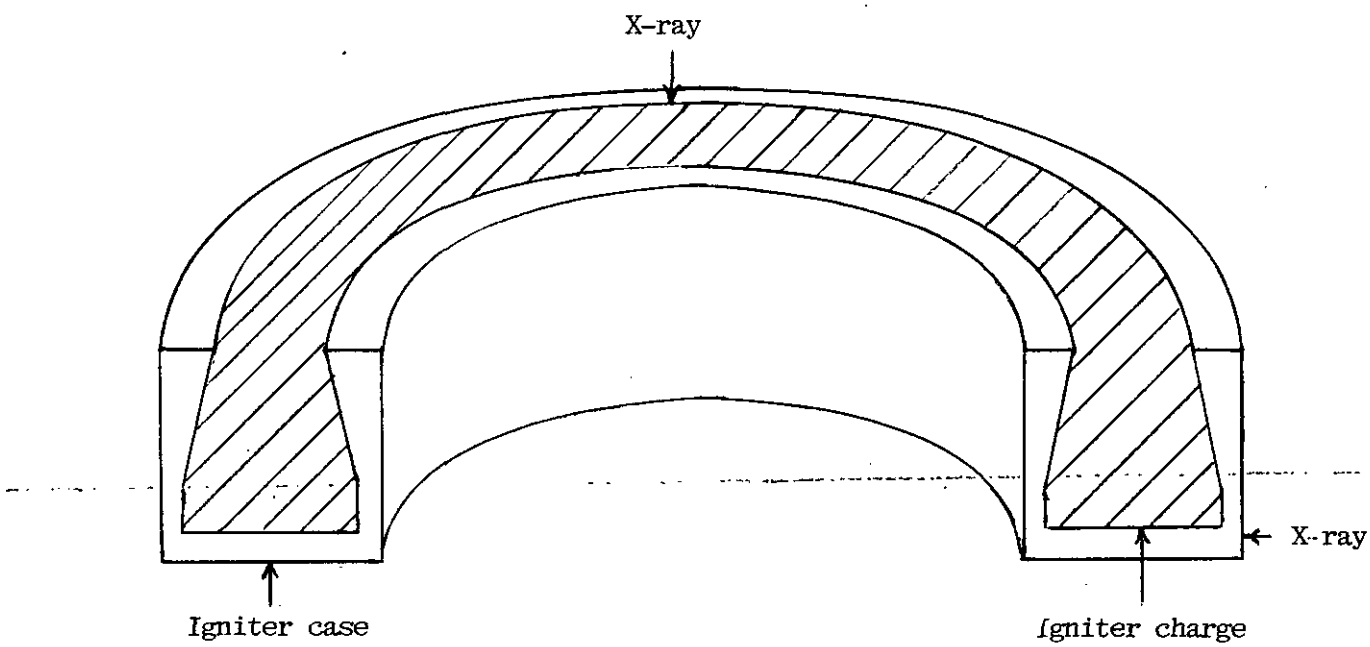


Figure 1. Radiographic Examination

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

3.7 Environmental requirements.

3.7.1 Temperature-humidity-altitude cycling. The igniters shall meet the requirements of 3.5, 3.6, and 3.8 after being tested in accordance with 4.5.3.

3.7.2 Shock. The igniters shall meet the requirements of 3.5, 3.6, and 3.8 after being tested in accordance with 4.5.4.

3.7.3 Vibration. The igniters shall meet the requirements of 3.5, 3.6, and 3.8 after being tested in accordance with 4.5.5. Unless otherwise specified in the contract, no resonant survey shall be conducted.

3.8 Performance requirements.

3.8.1 Closed bomb internal pressure. At -65 degrees Fahrenheit (°F), the igniters shall generate a minimum pressure of 600 pounds per square inch gage (psig) at 25 milliseconds (ms) and a maximum peak pressure of 1000 psig when tested in accordance with 4.5.6. At 165°F, the igniters shall generate a minimum pressure of 500 psig at 25 ms and a maximum peak pressure of 1000 psig when tested in accordance with 4.5.6.

3.8.2 Ballistics. The parameter values resulting from test firing the Rocket Catapults M9 and M37 Sustainers in accordance with 4.5.7 shall be within the requirements specified in Table 1.

TABLE I:

Test firing requirements for sustainers.

Performance requirements	Firing Temperatures			
	-65° F		165°F	
	Max.	Min.	Max.	Min.
Ignition delay (ms)	50	-	12	-
Maximum pressure (psig)	4600	-	4600	-
Action time (ms)	475	-	350	-
Pressure time integral (psig/sec)	-	550	-	550

3.9 Workmanship. All components shall be free of burrs and contaminants such as oil, grease, and dirt. The set screw and igniter case threads shall be free of ignition material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of the igniters shall be classified as follows:

a. Preproduction
b. Quality conformance

4.2.1 Preproduction inspection. Preproduction inspection shall consist of all the examinations and tests in the sequence specified in Table II and Figure 2. If any unit fails, the entire preproduction sample shall be rejected.

4.2.2 Quality conformance inspection. Quality conformance inspection shall consist of the following examinations and tests:

a. Weighing (4.5.1)
b. Visual and radiographic (4.5.2)
c. Closed bomb (4.5.6)

All production units shall be subjected to (a.) and (b.) and the failures rejected. The lot sample only (see 4.3.2) shall be subjected to the closed bomb test (c.).

4.3 Sampling.

4.3.1 Lot size. For the purpose of sampling, the production lot shall be not greater than 500 igniters.

4.3.2 Sample selection. Samples for quality conformance closed bomb tests shall be selected at random by the Government procuring activity after lot manufacture has been completed. The lot samples shall be as evenly divided as possible for test firings at -65°F and 165°F, Igniter samples shall not be preselected during production. Igniters selected for samples shall not be considered as part of the quantity specified for delivery by the contract. One ignition material production batch may be used in more than one loaded igniter production lot. The igniter sample size shall be divided as equally as possible among ignition material batches. Igniter samples shall be selected as follows:

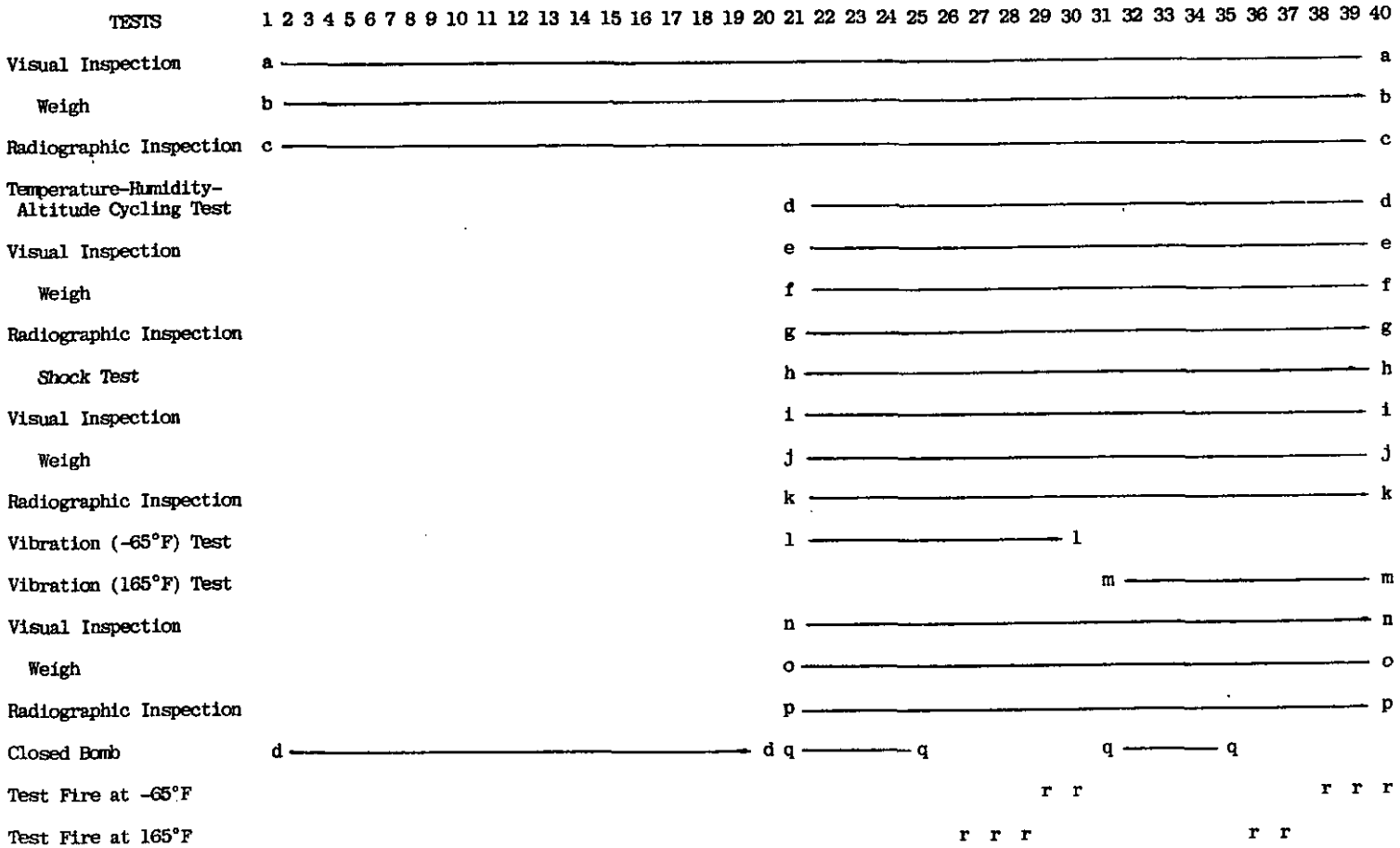
Lot size	Sample size
0-40	6
41-65	8
66-110	10
111-180	16
181-300	20
301-500	25

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

TABLE II.

Preproduction Test Schedule

Igniter Numbers



NOTE: Letters refer to testing sequence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

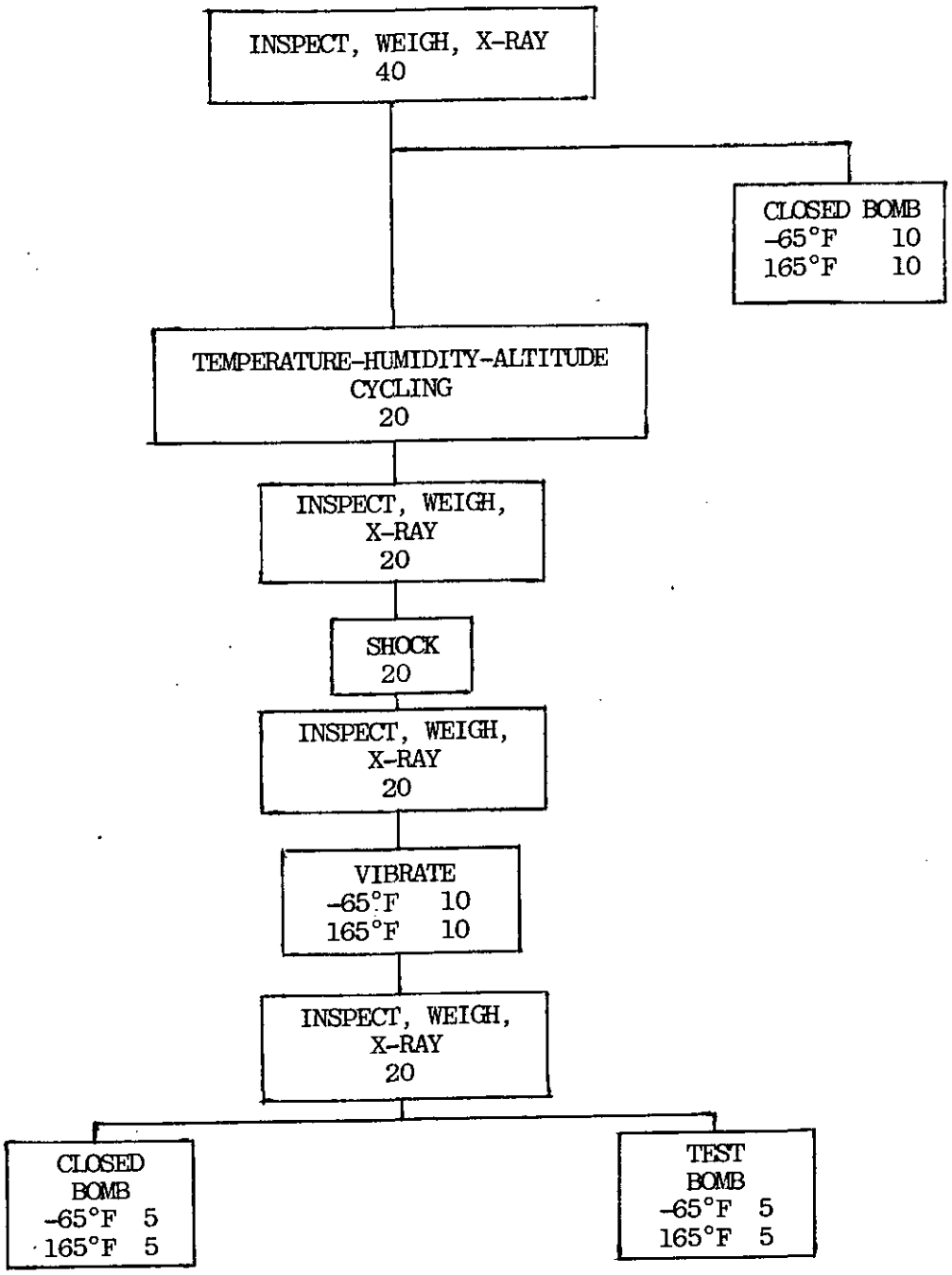


Figure 2. Preproduction test program

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

4.4 Test conditions. Unless otherwise specified, the igniters shall be tested under the following conditions:

a. 70° ± 5°F temperature
b. Less than 50 percent relative humidity
c. Ground elevation of test facility

4.5 Test methods.

4.5.1 Weighing. Each loaded igniter shall be weighed to the nearest milligram and the weight recorded. The weight of the igniter case, as provided by the contractor (3.5), shall be subtracted from the weight of the loaded igniter to determine that the weight of the ignition material conforms to Drawing 8596874. Any deviation from the requirements in the weight of the ignition material shall be cause for rejection of that igniter. Each igniter subjected to environmental testing (4.5.3, 4.5.4, and 5.5) shall be reweighed for conformance to 3.5. Nonconformance to 3.5 shall be cause for rejection of the igniter.

4.5.2 Visual and radiographic inspection. All igniters shall be visually examined and radiographically examined in accordance with MIL-STD-1166, level 2-IT. Radiographs shall be able to show defects 0.03 inch in length. Any igniter not meeting the requirements of 3.2, 3.6, 3.9 and section 5 shall be cause for rejection of the igniter. The igniters shall be reexamined after each of the environmental tests (4.5.3, 4.5.4, and 4.5.5). Any igniter not meeting the requirements of 3.6 shall be cause for rejection of the igniter. One set of radiographs shall be forwarded to the Naval Ordnance Station, Indian Head, Md. 20640 for approval.

4.5.3 Temperature-humidity-altitude cycling. The igniter assembly shall withstand temperature-humidity-altitude cycling in accordance with Method 518 of MIL-STD-810, Procedure 1.

4.5.4 Shock tests. The igniter assembly shall withstand shock in accordance with Method 516 of MIL-STD-810, Procedure 1, without evidence of any disintegration of the igniter charge. The shock pulse wave form shall be a terminal peak saw tooth. The peak amplitude shall be 20 G's and the duration shall be 11 ms.

4.5.5 Vibration tests. The igniter assembly shall be subjected to random vibration in accordance with MIL-STD-810, Method 514.2, "Random Vibration Test", for 6 hours, 2 hours along each axis. The samples shall be preconditioned as specified in Table II. The input acceleration shall be in accordance with MIL-STD-810, figure 514.2-2A, curve A, with a "Wo" factor of .1G^2/Hz. If the systems must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing shall be stopped and

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

the systems reconditioned. A reconditioning time of 2 minutes for every minute the systems are outside the specified temperature envelope shall be required. A continuous temperature history of the systems shall be kept for all conditioning and testing time. Unless otherwise noted, the systems shall be mounted on the test fixture, preloaded in a manner that will stimulate actual catapult installation and excited through the attachment fittings.

4.5.6 Closed bomb test. All igniter samples, temperature conditioned for 8 hours according to the applicable paragraph 4.2.1 or 4.3.2, shall be tested in a closed bomb in accordance with procedures previously approved by the Government procuring activity. If any igniter fails to meet the pressures specified in 3.8.1, the entire lot shall be rejected.

4.5.7 Test firing. Each igniter shall be loaded in a Rocket Catapult M9 or M37, temperature conditioned for 8 hours as specified in Table II, and test-fired at the conditioned temperature. If any igniter fails to meet the requirements of 3.8.2 the igniter lot shall be rejected.

4.6 Submission of data. When specified in the contract and at the time the igniters are submitted for testing, the supplier shall supply the following ignition material information, accompanied by a certificate which attests that the information is correct and applicable to the product submitted:

a. A statement that the ignition material complies with all requirements and quality assurance provisions of AS 2683.

b. Certificates of analyses including heat-of-explosion test and differential thermal analysis results.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A.

5.1.1 Level A. Each loaded igniter shall be wrapped first in 0.020-inch-thick aluminum foil conforming to QQ-A-1876 and then with sufficient cushioning material conforming to PPP-C-843, type III, class A. The wrapped loaded igniter shall be placed in a water vaporproof bag conforming to MIL-P-116, method 1A8, and sealed.

5.2 Packing. Packing shall be level A.

5.2.1 Level A. The bagged igniters shall be packed into Signal Containers Mk 3 Mod 0 conforming to LD 255173, and in accordance with 49 CFR 171-190.

5.3 Marking. In addition to the marking required by the contract or order, unit packages and signal containers shall be marked in accordance with the requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

5.3.1 Special marking. Each bagged igniter (see 5.1.1) shall be labeled with the following warning in 1/8-inch minimum high letters using black stencil ink conforming to TT-I-1795:

WARNING

This material is ignited from sparks, flame, or handling.

5.4 Data cards. Data cards, DD Form 1650 shall be furnished and filled out in accordance with MIL-STD-1167 by the supplier for each production lot of ignition material. Two data cards shall be placed in each shipping container. Additional data cards shall be provided as directed by the Government representative or the Government procuring activity.

6. NOTES

6.1 Intended use. The igniter is intended for use as an auxiliary ignition charge for rocket catapults, M9 and M37.

6.2 Ordering data. Procurement documents shall specify, but not be limited to, the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification
b. Preproduction sample if other than specified
c. Test facility responsible for preproduction testing
d. Invocation of MIL-Q-9858
e. Resonant frequency, if any (3.7.3)
f. Test conditions if other than specified (see 4.1, 4.4, & 4.5.5)
g. Safety precautions (see 6.3)

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.6	Certificate of Analysis	-
5.4	Ammunition Data Card	DI-E-2001

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

6.3 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 Warning. The ignition material, when dry, is a dangerous, highly sensitive explosive. All tests should be accomplished in a room set aside for work on primary explosives. Extreme precautions should be taken in all stages of work, and special care should be taken to avoid accumulation of static charge. Conductive containers should be used when handling the dry ignition material. No quantity of ignition material larger than the minimum quantity necessary should be taken at any one time.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Custodian:
NAVY-OS

Preparing activity:
NAVY-OS

Project No. 1377-N643

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-122/6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-I-60667A(OS), IGNITER, AUXILIARY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*.

4. REMARKS.

SUBMITTED BY *(Printed or typed name and address - Optional)*.

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED.

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:54Z
Check the source to verify that this is the current version before use.